

CCNA Project

Multi-Site Enterprise Network with
Static Routing, Web Servers & DNS

Sites: Mumbai-DC | Chennai-DC | Pune-DC | Global-DNS | Branch Office

1. Overall Network Topology

A multi-site enterprise WAN built in EVE-NG / GNS3, connecting four sites through routers R1-R7: Mumbai-DC (Linux1/Linux2/Linux3 on 10.10/10.20/10.30), Chennai-DC (Linux4/Linux6/Linux8 on 200.10.156.x), Pune-DC (IIS web server at 182.46.0.10), and a Global-DNS site (DNS server at 135.82.0.10). A remote branch with two Windows clients (Win1, Win2) connects via R7.

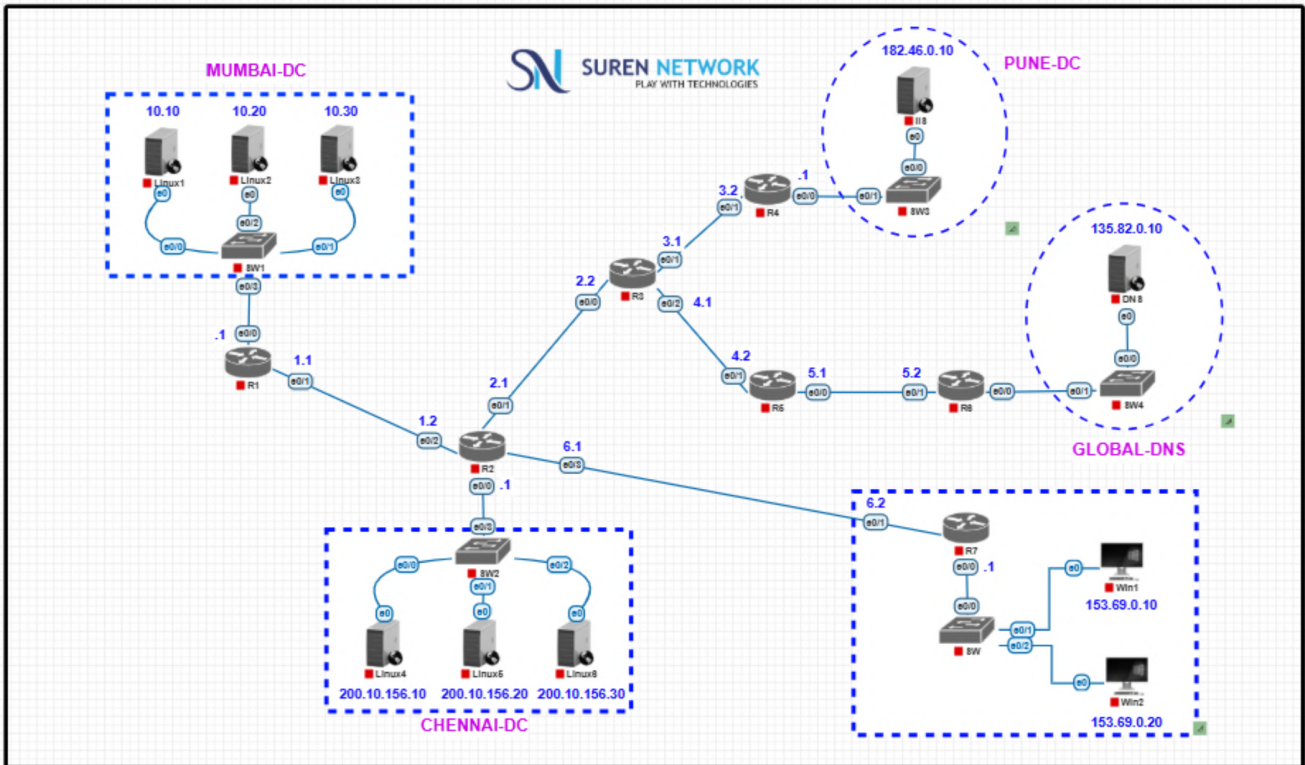


Fig. 1 - Full topology diagram showing all routers, switches, servers, and IP addressing.

2. Router R1 Configuration

R1 connects the Mumbai-DC LAN (10.0.0.1) to the WAN link toward R2 (1.0.0.1). The output of 'show ip int br' confirms both interfaces are up/up, and 'show ip route' shows static routes pointing to every remote subnet (2.0.0.0/8 through 6.0.0.0/8, plus the Pune, Global-DNS, Chennai and branch subnets) via next hop 1.0.0.2 (R2).

 R1

```
R1#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              10.0.0.1        YES manual  up          up
Ethernet0/1              1.0.0.1         YES manual  up          up
Ethernet0/2              unassigned      YES TFTP   up          up
Ethernet0/3              unassigned      YES TFTP   up          up
R1#
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       1.0.0.0/8 is directly connected, Ethernet0/1
L       1.0.0.1/32 is directly connected, Ethernet0/1
S       2.0.0.0/8 [1/0] via 1.0.0.2
S       3.0.0.0/8 [1/0] via 1.0.0.2
S       4.0.0.0/8 [1/0] via 1.0.0.2
S       5.0.0.0/8 [1/0] via 1.0.0.2
S       6.0.0.0/8 [1/0] via 1.0.0.2
    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       135.82.0.0/16 [1/0] via 1.0.0.2
S       153.69.0.0/16 [1/0] via 1.0.0.2
S       182.46.0.0/16 [1/0] via 1.0.0.2
S       200.10.156.0/24 [1/0] via 1.0.0.2
R1#
```

Fig. 2 - R1 interface status and static routing table.

3. Router R3 Configuration

R3 sits at the core of the topology, with three active interfaces: Ethernet0/0 (2.0.0.2) toward R2, Ethernet0/1 (3.0.0.1) toward R4/Pune-DC, and Ethernet0/2 (4.0.0.1) toward R6/Global-DNS. Its routing table shows static routes reaching every other subnet in the network through the appropriate next hop.

 R3

```
R3#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              2.0.0.2         YES manual up          up
Ethernet0/1              3.0.0.1         YES manual up          up
Ethernet0/2              4.0.0.1         YES manual up          up
Ethernet0/3              unassigned      YES TFTP  up          up
R3#
R3#
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 2.0.0.1
S    2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    2.0.0.0/8 is directly connected, Ethernet0/0
L    2.0.0.2/32 is directly connected, Ethernet0/0
S    3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    3.0.0.0/8 is directly connected, Ethernet0/1
L    3.0.0.1/32 is directly connected, Ethernet0/1
S    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    4.0.0.0/8 is directly connected, Ethernet0/2
L    4.0.0.1/32 is directly connected, Ethernet0/2
S    5.0.0.0/8 [1/0] via 4.0.0.2
S    6.0.0.0/8 [1/0] via 2.0.0.1
S    10.0.0.0/8 [1/0] via 2.0.0.1
S    135.82.0.0/16 [1/0] via 4.0.0.2
S    153.69.0.0/16 [1/0] via 2.0.0.1
S    182.46.0.0/16 [1/0] via 3.0.0.2
S    200.10.156.0/24 [1/0] via 2.0.0.1
R3#
R3#
```

Fig. 3 - R3 interface status and static routing table.

4. Router R2 Configuration

R2 is the central hub router with four active interfaces: Ethernet0/0 (200.10.156.1) to Chennai-DC, Ethernet0/1 (2.0.0.1) to R3, Ethernet0/2 (1.0.0.2) to R1/Mumbai-DC, and Ethernet0/3 (6.0.0.1) to R7/branch office. The routing table confirms reachability to all remaining subnets.

 R2

```
R2#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              200.10.156.1   YES manual up          up
Ethernet0/1              2.0.0.1        YES manual up          up
Ethernet0/2              1.0.0.2        YES manual up          up
Ethernet0/3              6.0.0.1        YES manual up          up
R2#
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       1.0.0.0/8 is directly connected, Ethernet0/2
L       1.0.0.2/32 is directly connected, Ethernet0/2
    2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       2.0.0.0/8 is directly connected, Ethernet0/1
L       2.0.0.1/32 is directly connected, Ethernet0/1
S       3.0.0.0/8 [1/0] via 2.0.0.2
S       4.0.0.0/8 [1/0] via 2.0.0.2
S       5.0.0.0/8 [1/0] via 2.0.0.2
    6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       6.0.0.0/8 is directly connected, Ethernet0/3
L       6.0.0.1/32 is directly connected, Ethernet0/3
S      10.0.0.0/8 [1/0] via 1.0.0.1
S     135.82.0.0/16 [1/0] via 2.0.0.2
S     153.69.0.0/16 [1/0] via 6.0.0.2
S     182.46.0.0/16 [1/0] via 2.0.0.2
    200.10.156.0/24 is variably subnetted, 2 subnets, 2 masks
C     200.10.156.0/24 is directly connected, Ethernet0/0
L     200.10.156.1/32 is directly connected, Ethernet0/0
R2#
```

Fig. 4 - R2 interface status and static routing table.

5. Router R4 Configuration

R4 links the core network to the Pune-DC site. Ethernet0/0 (182.46.0.1) faces the Pune IIS server/switch, while Ethernet0/1 (3.0.0.2) connects back to R3. All remote networks are reachable via static routes pointing to 3.0.0.1.

 R4

```
R4#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              182.46.0.1     YES manual  up          up
Ethernet0/1              3.0.0.2        YES manual  up          up
Ethernet0/2              unassigned     YES TFTP   up          up
Ethernet0/3              unassigned     YES TFTP   up          up
R4#
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 3.0.0.1
S    2.0.0.0/8 [1/0] via 3.0.0.1
     3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     3.0.0.0/8 is directly connected, Ethernet0/1
L     3.0.0.2/32 is directly connected, Ethernet0/1
S    4.0.0.0/8 [1/0] via 3.0.0.1
S    5.0.0.0/8 [1/0] via 3.0.0.1
S    6.0.0.0/8 [1/0] via 3.0.0.1
S   10.0.0.0/8 [1/0] via 3.0.0.1
S   135.82.0.0/16 [1/0] via 3.0.0.1
S   153.69.0.0/16 [1/0] via 3.0.0.1
     182.46.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     182.46.0.0/16 is directly connected, Ethernet0/0
L     182.46.0.1/32 is directly connected, Ethernet0/0
S   200.10.156.0/24 [1/0] via 3.0.0.1
R4#
R4#
```

Fig. 5 - R4 interface status and static routing table.

6. Router R6 Configuration

R6 connects toward the Global-DNS site. Ethernet0/0 (135.82.0.1) faces the DNS server subnet, and Ethernet0/1 (5.0.0.2) connects to R5. Static routes provide reachability to every other site in the network via 5.0.0.1.

 R6

```
R6#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              135.82.0.1      YES manual  up          up
Ethernet0/1              5.0.0.2         YES manual  up          up
Ethernet0/2              unassigned      YES TFTP   up          up
Ethernet0/3              unassigned      YES TFTP   up          up
```

R6#

```
R6#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 5.0.0.1
S    2.0.0.0/8 [1/0] via 5.0.0.1
S    3.0.0.0/8 [1/0] via 5.0.0.1
S    4.0.0.0/8 [1/0] via 5.0.0.1
    5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    5.0.0.0/8 is directly connected, Ethernet0/1
L    5.0.0.2/32 is directly connected, Ethernet0/1
S    6.0.0.0/8 [1/0] via 5.0.0.1
S    10.0.0.0/8 [1/0] via 5.0.0.1
C    135.0.0.0/8 is directly connected, Ethernet0/0
    135.82.0.0/32 is subnetted, 1 subnets
L    135.82.0.1 is directly connected, Ethernet0/0
S    153.69.0.0/16 [1/0] via 5.0.0.1
S    182.46.0.0/16 [1/0] via 5.0.0.1
S    200.10.156.0/24 [1/0] via 5.0.0.1
```

R6#

Fig. 6 - R6 interface status and static routing table.

7. Router R5 Configuration

R5 is a transit router between R6 (Global-DNS side) and R3 (core). Ethernet0/0 (5.0.0.1) faces R6, and Ethernet0/1 (4.0.0.2) faces R3. Its routing table shows static routes to all remote subnets in both directions.

 R5

```
R5#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              5.0.0.1        YES manual  up          up
Ethernet0/1              4.0.0.2        YES manual  up          up
Ethernet0/2              unassigned     YES TFTP   up          up
Ethernet0/3              unassigned     YES TFTP   up          up
```

R5#

```
R5#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 4.0.0.1
S    2.0.0.0/8 [1/0] via 4.0.0.1
S    3.0.0.0/8 [1/0] via 4.0.0.1
    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    4.0.0.0/8 is directly connected, Ethernet0/1
L    4.0.0.2/32 is directly connected, Ethernet0/1
    5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    5.0.0.0/8 is directly connected, Ethernet0/0
L    5.0.0.1/32 is directly connected, Ethernet0/0
S    6.0.0.0/8 [1/0] via 4.0.0.1
S    10.0.0.0/8 [1/0] via 4.0.0.1
S    135.82.0.0/16 [1/0] via 5.0.0.2
S    153.69.0.0/16 [1/0] via 4.0.0.1
S    182.46.0.0/16 [1/0] via 4.0.0.1
S    200.10.156.0/24 [1/0] via 4.0.0.1
```

R5#

R5#

Fig. 7 - R5 interface status and static routing table.

8. Router R7 Configuration

R7 connects the core network to the remote branch office. Ethernet0/0 (153.69.0.1) faces the branch LAN switch (Win1/Win2), and Ethernet0/1 (6.0.0.2) connects back to R2. All other site subnets are reachable via static routes through 6.0.0.1.

 R7

```
R7#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              153.69.0.1      YES manual up          up
Ethernet0/1              6.0.0.2         YES manual up          up
Ethernet0/2              unassigned      YES TFTP  up          up
Ethernet0/3              unassigned      YES TFTP  up          up
R7#
R7#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    1.0.0.0/8 [1/0] via 6.0.0.1
S    2.0.0.0/8 [1/0] via 6.0.0.1
S    3.0.0.0/8 [1/0] via 6.0.0.1
S    4.0.0.0/8 [1/0] via 6.0.0.1
S    5.0.0.0/8 [1/0] via 6.0.0.1
     6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    6.0.0.0/8 is directly connected, Ethernet0/1
L    6.0.0.2/32 is directly connected, Ethernet0/1
S    10.0.0.0/8 [1/0] via 6.0.0.1
S    135.82.0.0/16 [1/0] via 6.0.0.1
     153.69.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    153.69.0.0/16 is directly connected, Ethernet0/0
L    153.69.0.1/32 is directly connected, Ethernet0/0
S    182.46.0.0/16 [1/0] via 6.0.0.1
S    200.10.156.0/24 [1/0] via 6.0.0.1
R7#
```

Fig. 8 - R7 interface status and static routing table.

9. Branch Client (Win1) IP Configuration

The Windows XP client Win1 at the branch office is configured with a static IP address 153.69.0.10, subnet mask 255.255.0.0, and default gateway 153.69.0.1 (R7), via the TCP/IP Properties dialog.

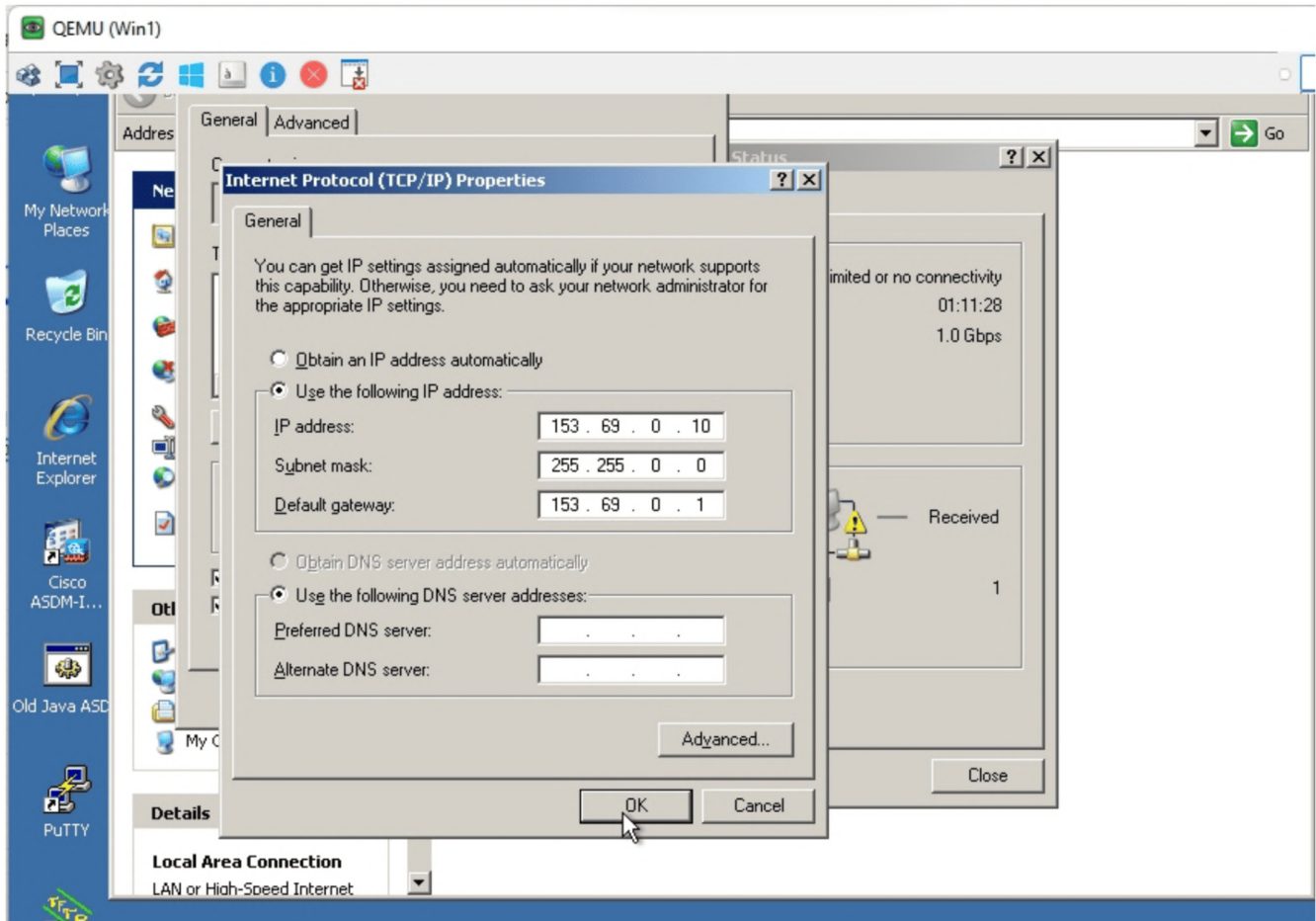
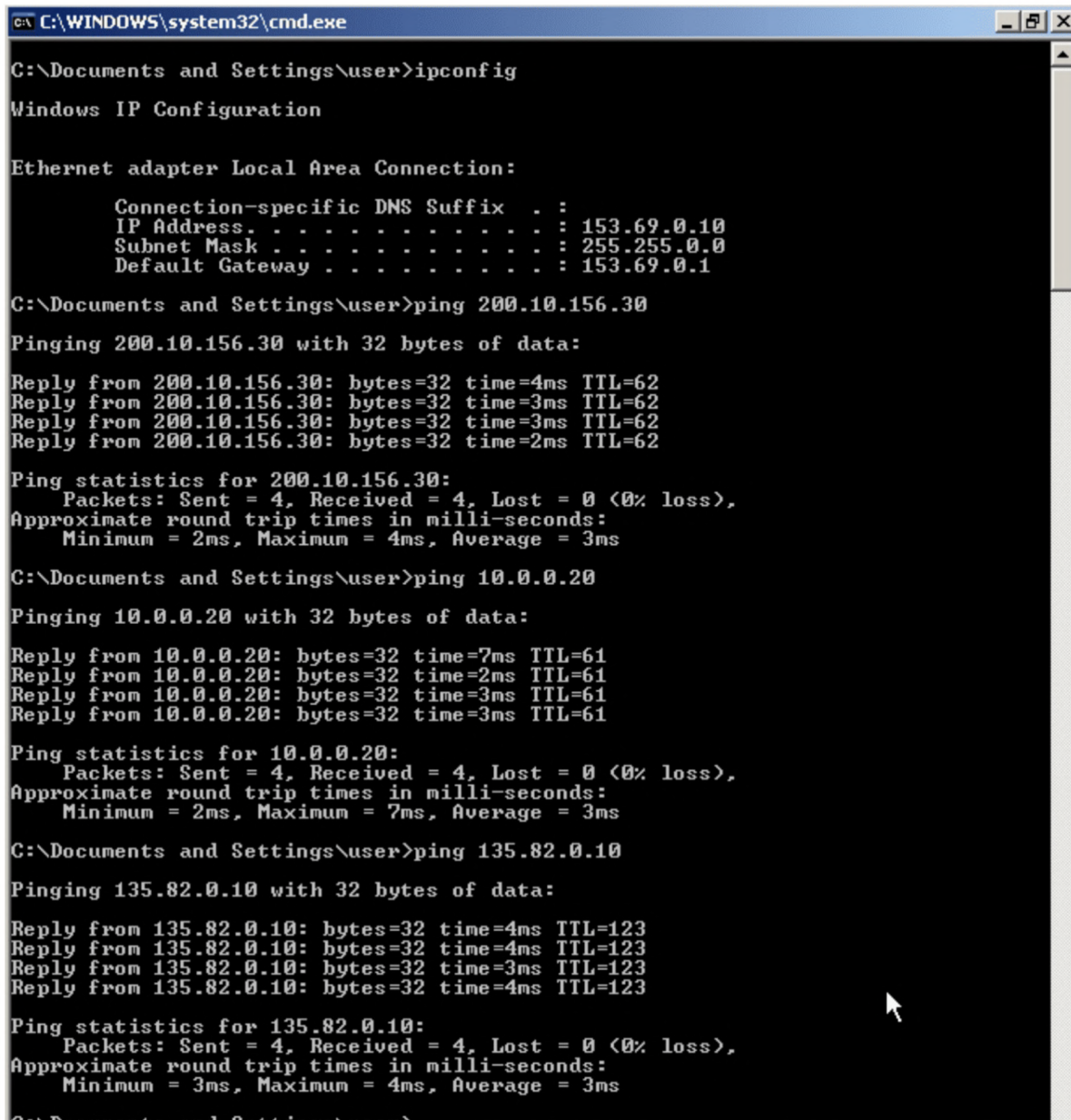


Fig. 9 - TCP/IP configuration on Win1.

10. Win1 Connectivity Test - Part 1

From Win1 (153.69.0.10), 'ipconfig' confirms the assigned address, and successive pings to 200.10.156.30 (Chennai-DC), 10.0.0.20 (Mumbai-DC), and 135.82.0.10 (Global-DNS) all succeed with 0% packet loss, proving end-to-end reachability across the WAN.

A screenshot of a Windows Command Prompt window titled "C:\WINDOWS\system32\cmd.exe". The window shows the output of the 'ipconfig' command, which displays the IP configuration for the 'Ethernet adapter Local Area Connection'. The IP address is 153.69.0.10, the subnet mask is 255.255.0.0, and the default gateway is 153.69.0.1. Following this, three ping commands are executed: 'ping 200.10.156.30', 'ping 10.0.0.20', and 'ping 135.82.0.10'. Each ping command shows four successful replies with 0% packet loss and provides statistics on the number of packets sent and received, as well as the approximate round trip times in milliseconds. The ping to 200.10.156.30 shows times of 4ms, 3ms, 3ms, and 2ms. The ping to 10.0.0.20 shows times of 7ms, 2ms, 3ms, and 3ms. The ping to 135.82.0.10 shows times of 4ms, 4ms, 3ms, and 4ms. The window is set against a black background with white text.

```
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 153.69.0.10
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 153.69.0.1

C:\Documents and Settings\user>ping 200.10.156.30

Pinging 200.10.156.30 with 32 bytes of data:

Reply from 200.10.156.30: bytes=32 time=4ms TTL=62
Reply from 200.10.156.30: bytes=32 time=3ms TTL=62
Reply from 200.10.156.30: bytes=32 time=3ms TTL=62
Reply from 200.10.156.30: bytes=32 time=2ms TTL=62

Ping statistics for 200.10.156.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\Documents and Settings\user>ping 10.0.0.20

Pinging 10.0.0.20 with 32 bytes of data:

Reply from 10.0.0.20: bytes=32 time=7ms TTL=61
Reply from 10.0.0.20: bytes=32 time=2ms TTL=61
Reply from 10.0.0.20: bytes=32 time=3ms TTL=61
Reply from 10.0.0.20: bytes=32 time=3ms TTL=61

Ping statistics for 10.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 7ms, Average = 3ms

C:\Documents and Settings\user>ping 135.82.0.10

Pinging 135.82.0.10 with 32 bytes of data:

Reply from 135.82.0.10: bytes=32 time=4ms TTL=123
Reply from 135.82.0.10: bytes=32 time=4ms TTL=123
Reply from 135.82.0.10: bytes=32 time=3ms TTL=123
Reply from 135.82.0.10: bytes=32 time=4ms TTL=123

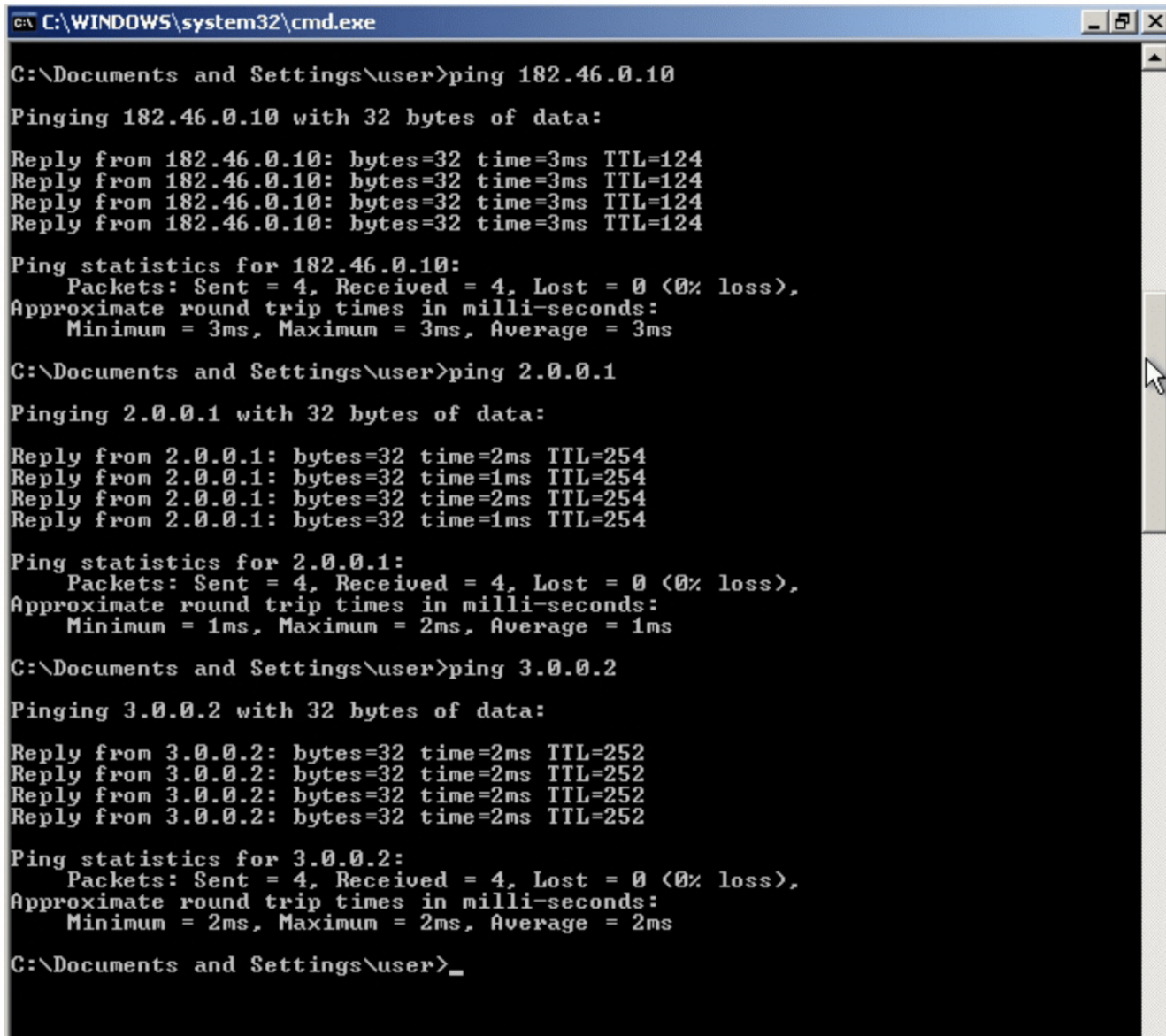
Ping statistics for 135.82.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\Documents and Settings\user>
```

Fig. 10 - ipconfig and ping tests to remote data centers from Win1.

11. Win1 Connectivity Test - Part 2

Further ping tests from Win1 to 182.46.0.10 (Pune-DC), 2.0.0.1 (R3), and 3.0.0.2 (R4) all return successful replies, confirming the branch office has full reachability to every site and core router in the topology.



```
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ping 182.46.0.10

Pinging 182.46.0.10 with 32 bytes of data:

Reply from 182.46.0.10: bytes=32 time=3ms TTL=124
Reply from 182.46.0.10: bytes=32 time=3ms TTL=124
Reply from 182.46.0.10: bytes=32 time=3ms TTL=124
Reply from 182.46.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 182.46.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 3ms, Average = 3ms

C:\Documents and Settings\user>ping 2.0.0.1

Pinging 2.0.0.1 with 32 bytes of data:

Reply from 2.0.0.1: bytes=32 time=2ms TTL=254
Reply from 2.0.0.1: bytes=32 time=1ms TTL=254
Reply from 2.0.0.1: bytes=32 time=2ms TTL=254
Reply from 2.0.0.1: bytes=32 time=1ms TTL=254

Ping statistics for 2.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 3.0.0.2

Pinging 3.0.0.2 with 32 bytes of data:

Reply from 3.0.0.2: bytes=32 time=2ms TTL=252
Reply from 3.0.0.2: bytes=32 time=2ms TTL=252
Reply from 3.0.0.2: bytes=32 time=2ms TTL=252
Reply from 3.0.0.2: bytes=32 time=2ms TTL=252

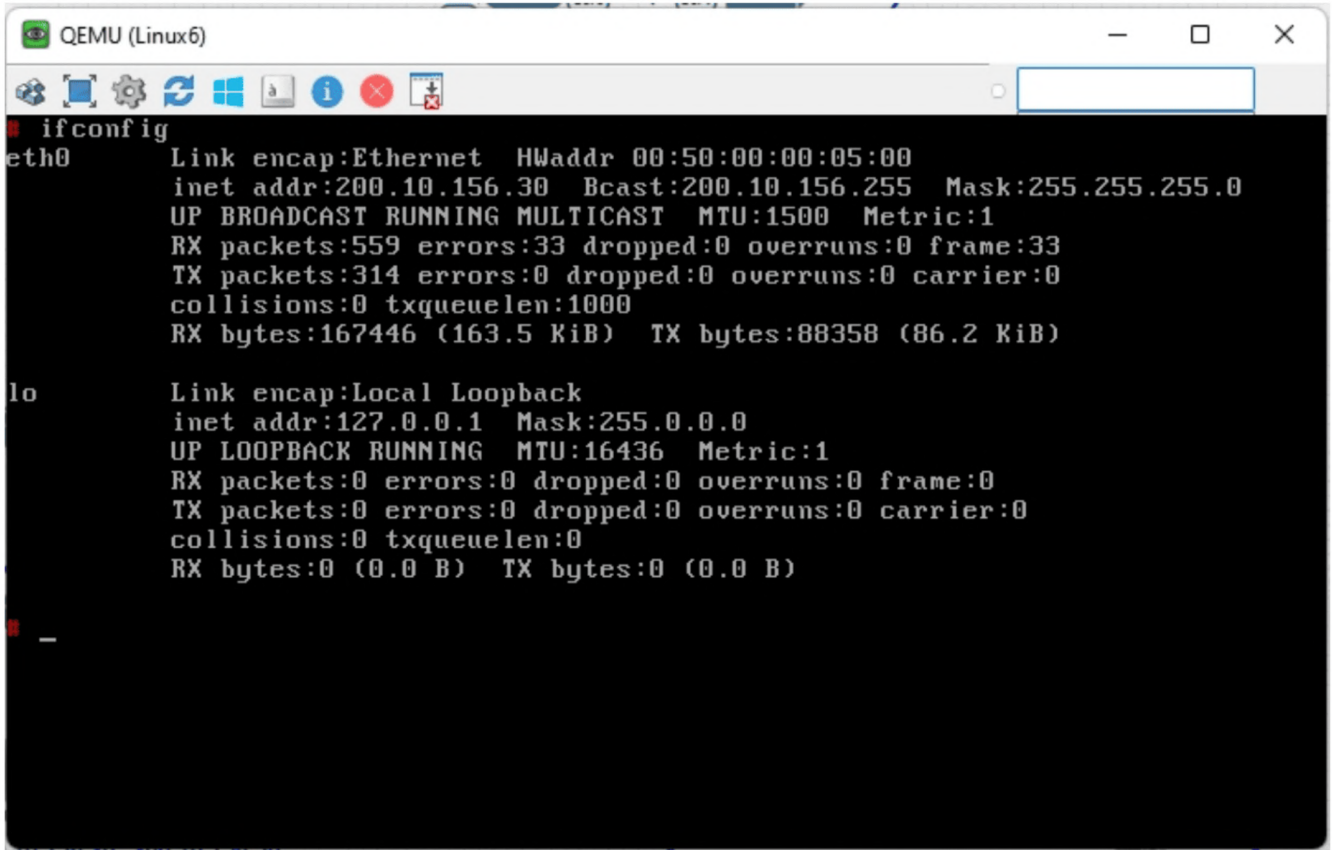
Ping statistics for 3.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>
```

Fig. 11 - Additional ping tests confirming full network reachability.

12. Chennai-DC Server (Linux6) Interface

The 'ifconfig' output on Linux6 in Chennai-DC shows eth0 configured with IP address 200.10.156.30/24, confirming it is correctly addressed on the Chennai LAN segment.



```
QEMU (Linux6)
ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:05:00
          inet addr:200.10.156.30  Bcast:200.10.156.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:559 errors:33 dropped:0 overruns:0 frame:33
          TX packets:314 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:167446 (163.5 KiB)  TX bytes:88358 (86.2 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

Fig. 12 - Network interface configuration on the Chennai-DC Linux server.

13. Chennai-DC Web Page Source

A simple HTML page is being served from Linux6 (200.10.156.30) via a lightweight web server (thttpd), identifying the page as loading from the Chennai Data Center.

A screenshot of a QEMU Linux64 virtual machine terminal window. The title bar reads "QEMU (Linux6)". Below the title bar is a standard Linux desktop taskbar with icons for file manager, settings, network, and other utilities. The terminal itself has a black background with white text. It displays the following HTML code:

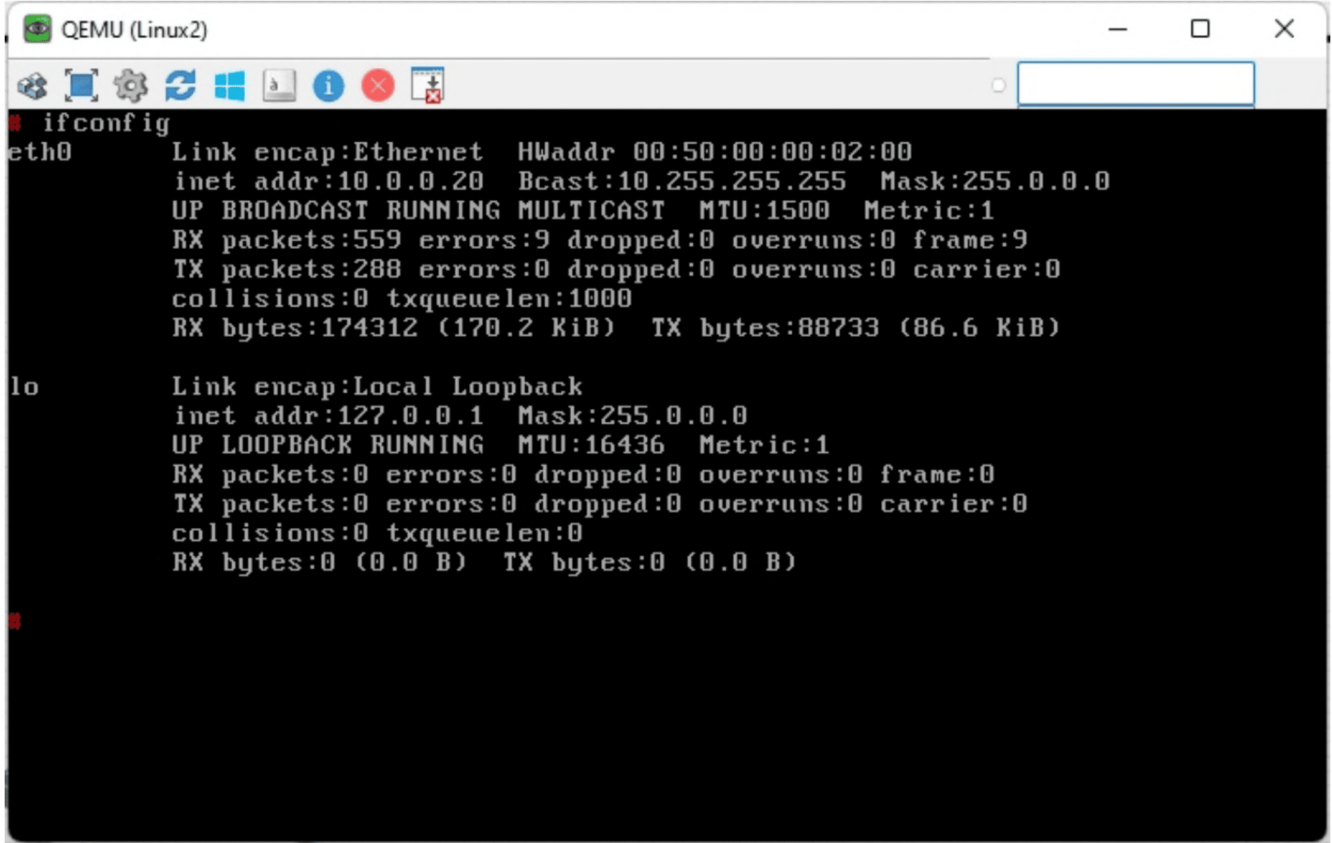
```
<h1><b>Hello everyone!</h1></b>  
<b><h1>This page is load from the server 200.10.156.30</h1></b>  
<h2 style=color:green><i>Chennai Data Center</i></h2>
```

The first two lines are followed by several tilde (~) characters, likely representing newlines or continuation of the document. At the bottom of the terminal, a status bar shows the path "/home/thttpd/index.html" and progress information "1/3 33%".

Fig. 13 - index.html content for the Chennai-DC web server.

14. Mumbai-DC Server (Linux2) Interface

The 'ifconfig' output on Linux2 in Mumbai-DC shows eth0 configured with IP address 10.0.0.20/8, confirming correct addressing on the Mumbai LAN segment.



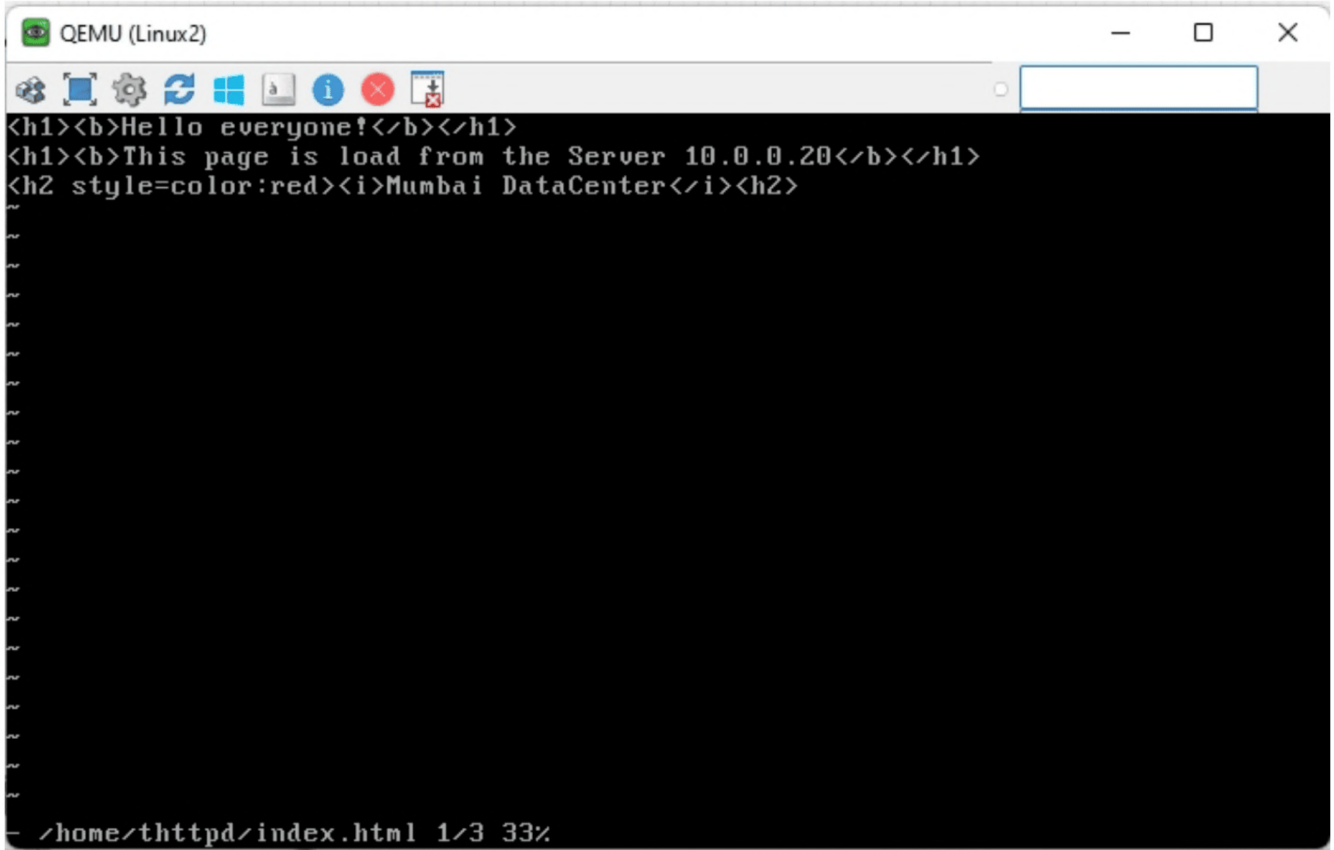
```
ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:02:00
          inet addr:10.0.0.20  Bcast:10.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:559 errors:9 dropped:0 overruns:0 frame:9
          TX packets:288 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:174312 (170.2 KiB)  TX bytes:88733 (86.6 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

Fig. 14 - Network interface configuration on the Mumbai-DC Linux server.

15. Mumbai-DC Web Page Source

The corresponding HTML page served from Linux2 (10.0.0.20) identifies itself as being loaded from the Mumbai Data Center.



```
QEMU (Linux2)
<h1><b>Hello everyone!</b></h1>
<h1><b>This page is load from the Server 10.0.0.20</b></h1>
<h2 style=color:red><i>Mumbai DataCenter</i></h2>

/home/tthttpd/index.html 1/3 33%
```

Fig. 15 - index.html content for the Mumbai-DC web server.

16. Pune-DC: Windows Server Manager Dashboard

The Pune-DC server (IIS) is running Windows Server 2016. The Server Manager Dashboard is opened as the starting point for installing the Web Server (IIS) role.

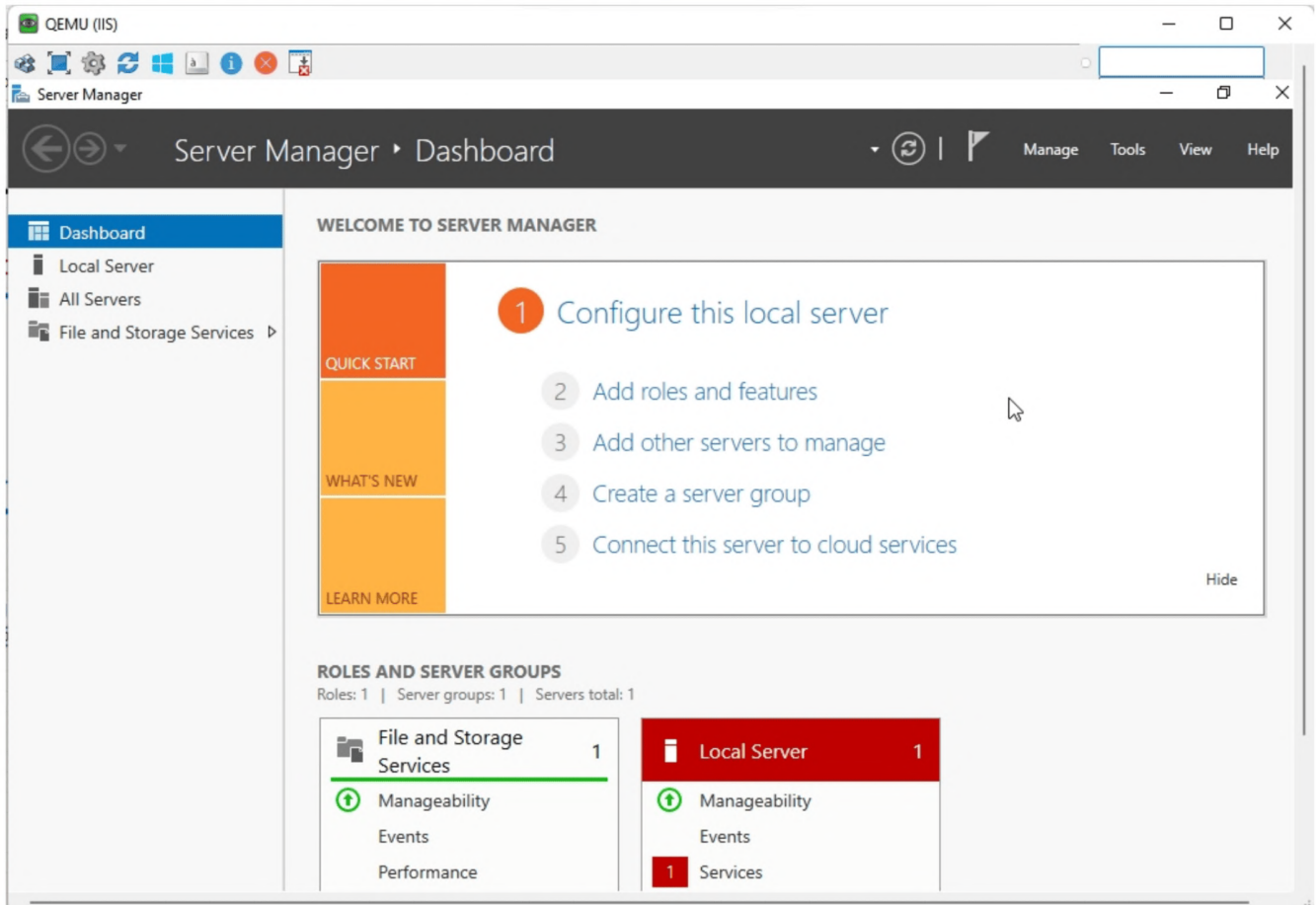


Fig. 16 - Server Manager Dashboard on the Pune-DC IIS server.

17. Pune-DC: IIS Role Installation Complete

The Add Roles and Features Wizard confirms that the Web Server (IIS) role, along with the IIS Management Console and common HTTP features, has been installed successfully on the Pune-DC server.

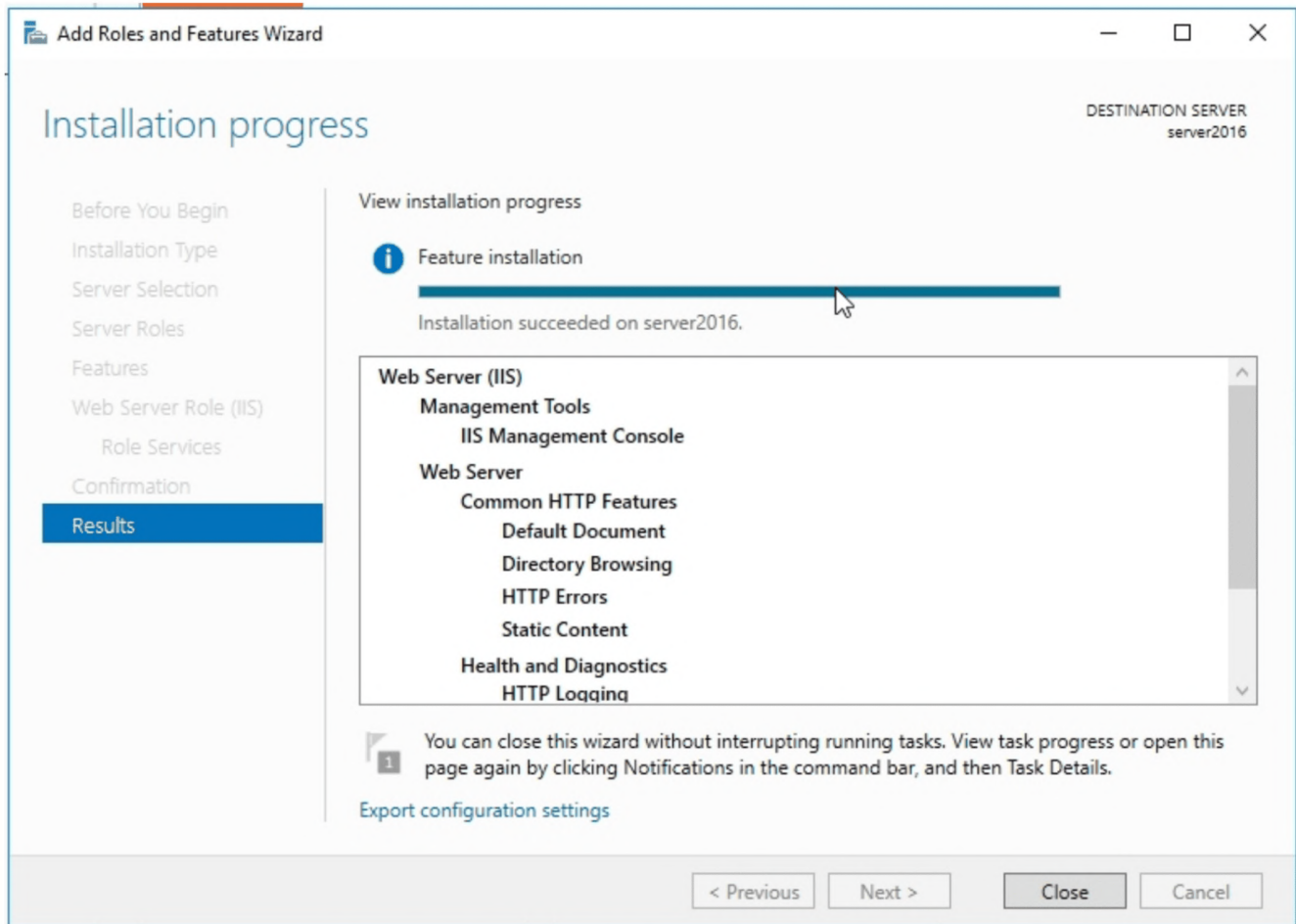


Fig. 17 - Installation progress showing successful IIS role installation.

18. Pune-DC: Static IP Configuration

The Pune-DC IIS server's network adapter is configured with static IP address 182.46.0.10, subnet mask 255.255.0.0, and default gateway 182.46.0.1 via IPv4 Properties.

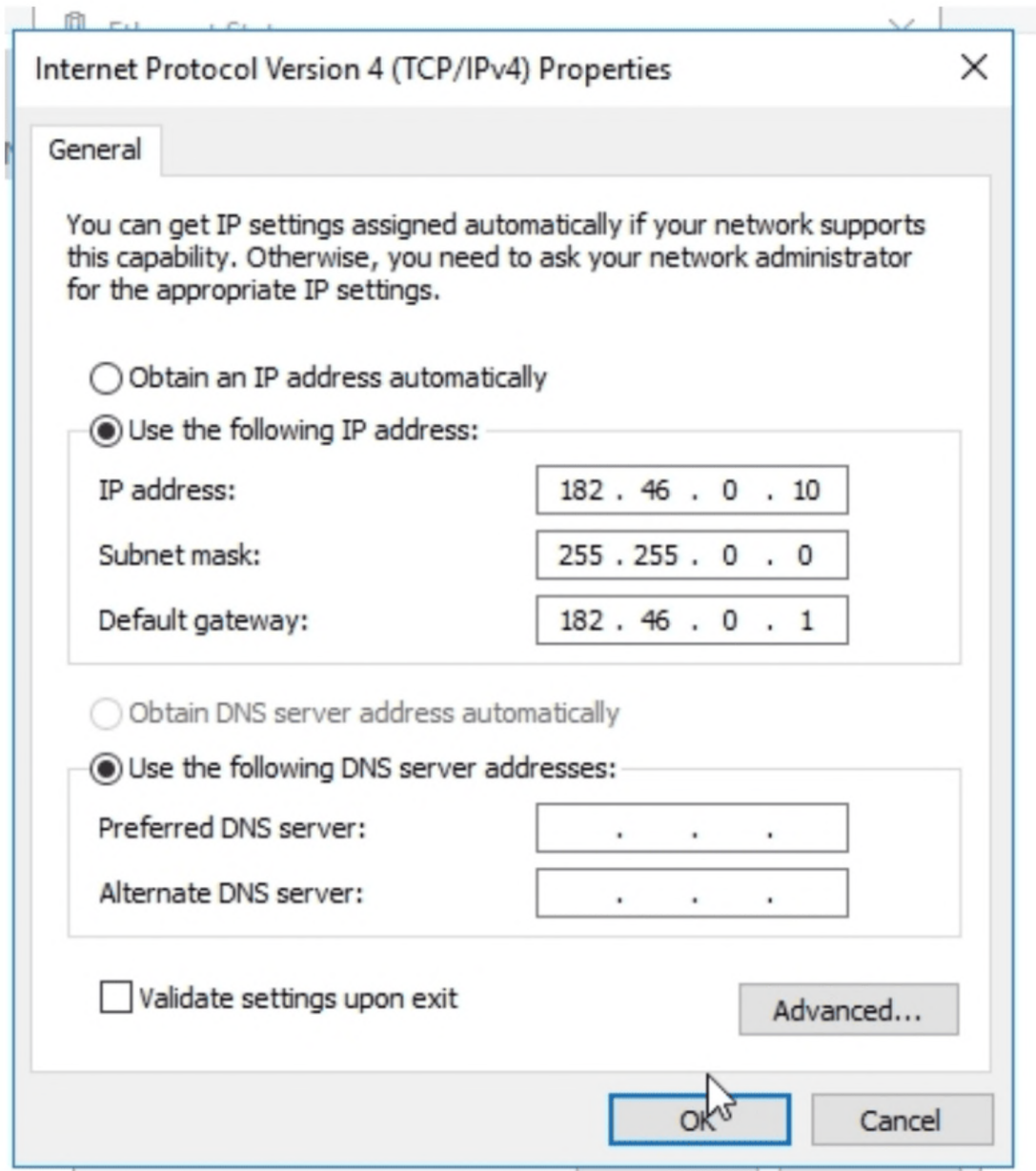


Fig. 18 - TCP/IPv4 configuration on the Pune-DC IIS server.

19. Global-DNS: Windows Server Manager Dashboard

The Global-DNS site's Windows Server 2016 machine is opened at the Server Manager Dashboard, ready to add the DNS Server role.

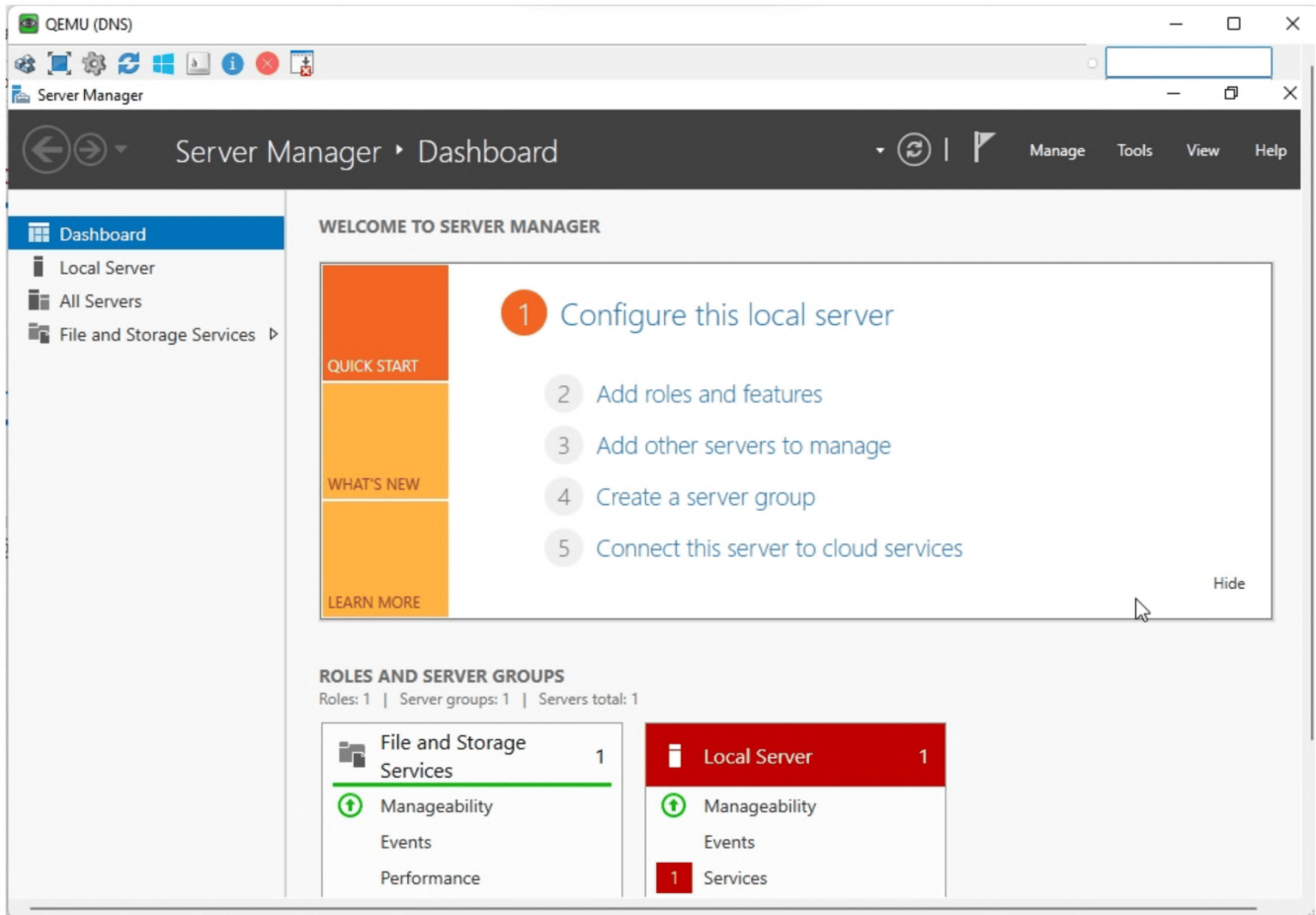


Fig. 19 - Server Manager Dashboard on the Global-DNS server.

20. Global-DNS: DNS Role Installation Complete

The Add Roles and Features Wizard confirms that the DNS Server role and DNS Server Tools have been installed successfully, enabling this machine to host DNS zones for the organization.

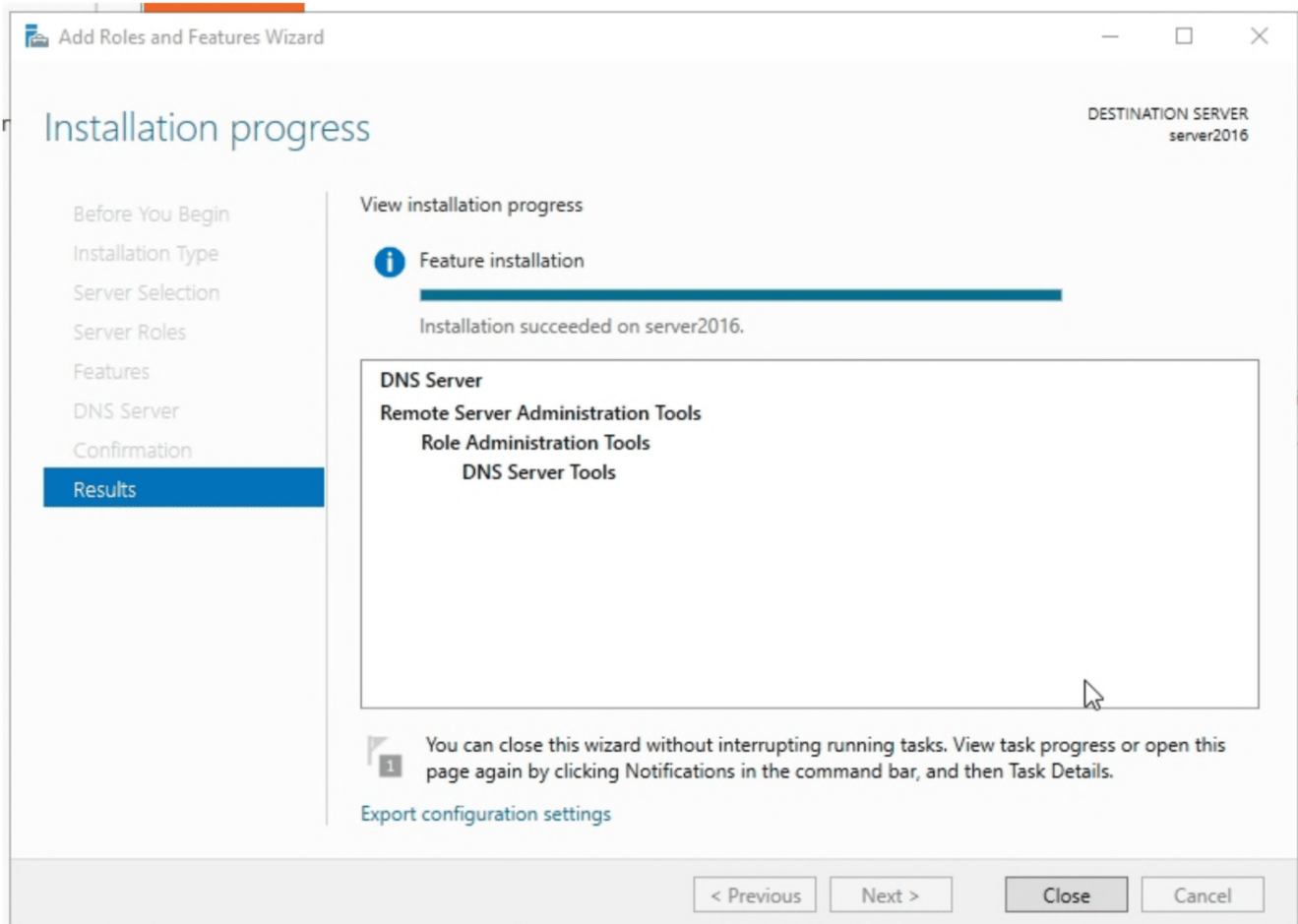


Fig. 20 - Installation progress showing successful DNS Server role installation.

21. Chennai-DC Verified by IP Address

From the branch client's browser, navigating to <http://200.10.156.30> successfully loads the Chennai-DC web page, confirming the Chennai web server is reachable and serving content across the WAN.

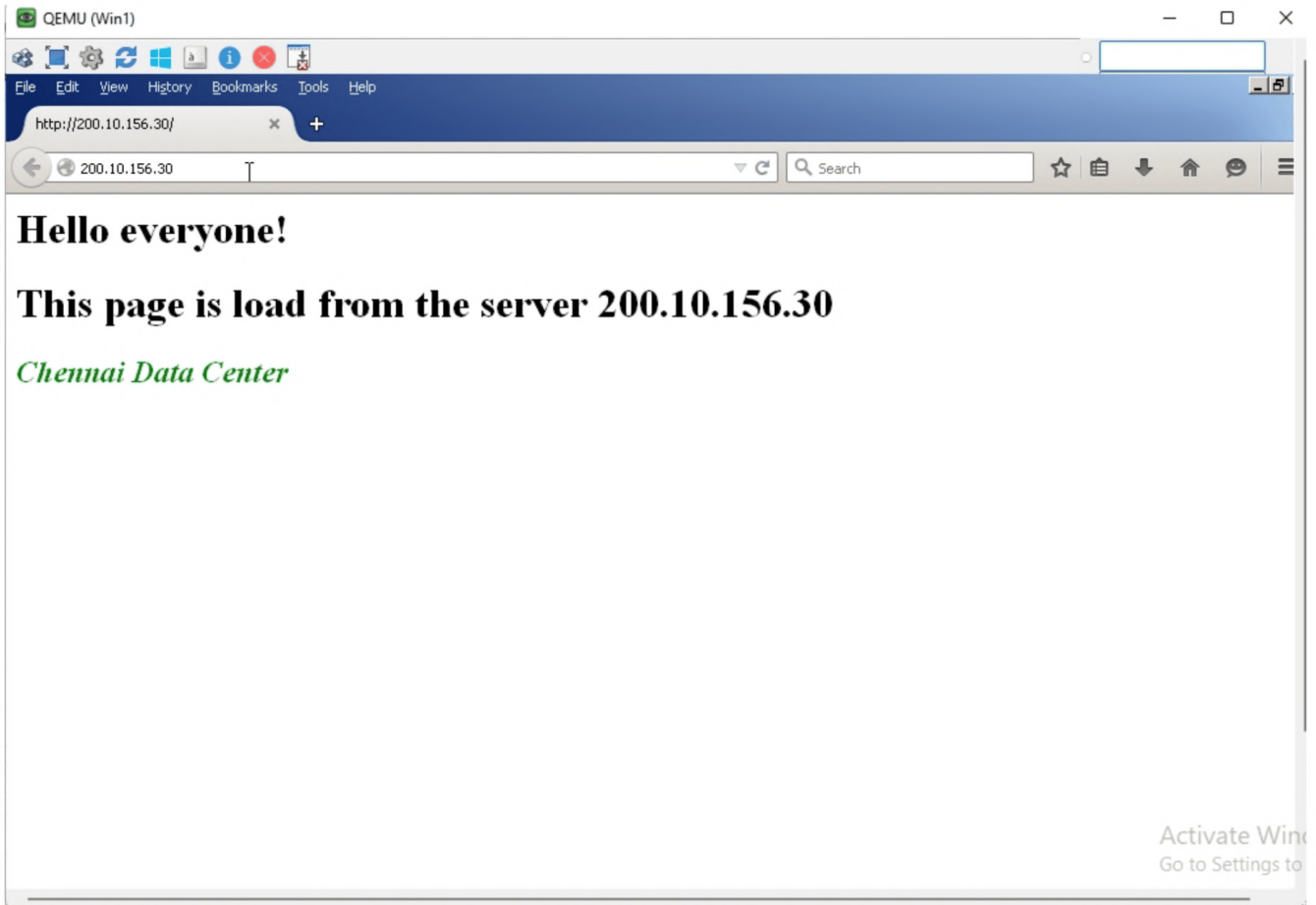


Fig. 21 - Browser test confirming Chennai-DC web server via IP address.

22. Pune-DC Verified by IP Address

Navigating to <http://182.46.0.10> loads the default IIS Windows Server welcome page, confirming the Pune-DC IIS installation is active and reachable from the branch office.

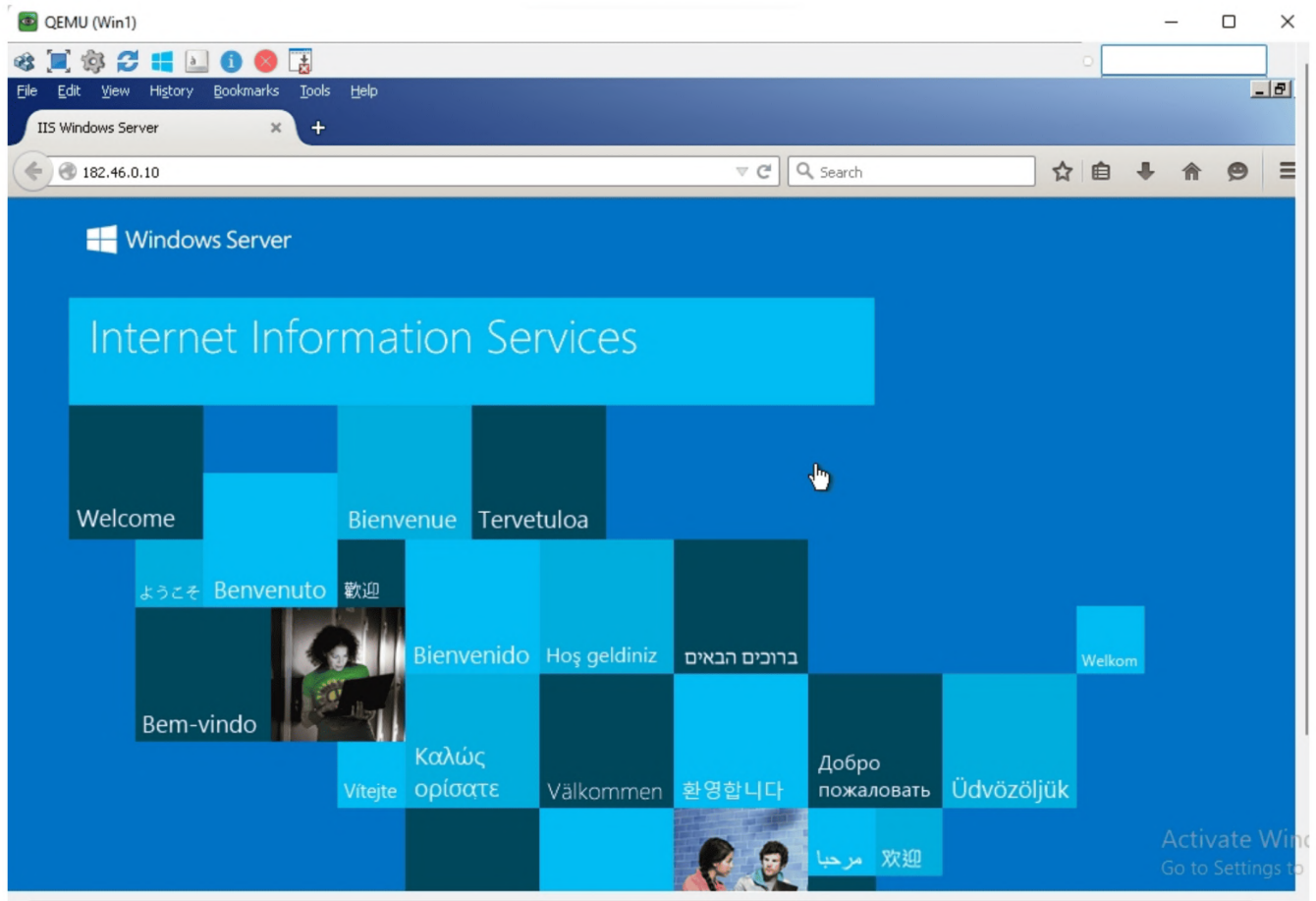


Fig. 22 - Browser test confirming Pune-DC IIS server via IP address.

23. Mumbai-DC Verified by IP Address

Navigating to <http://10.0.0.20> successfully loads the Mumbai-DC web page, confirming end-to-end reachability to the Mumbai data center's web server.

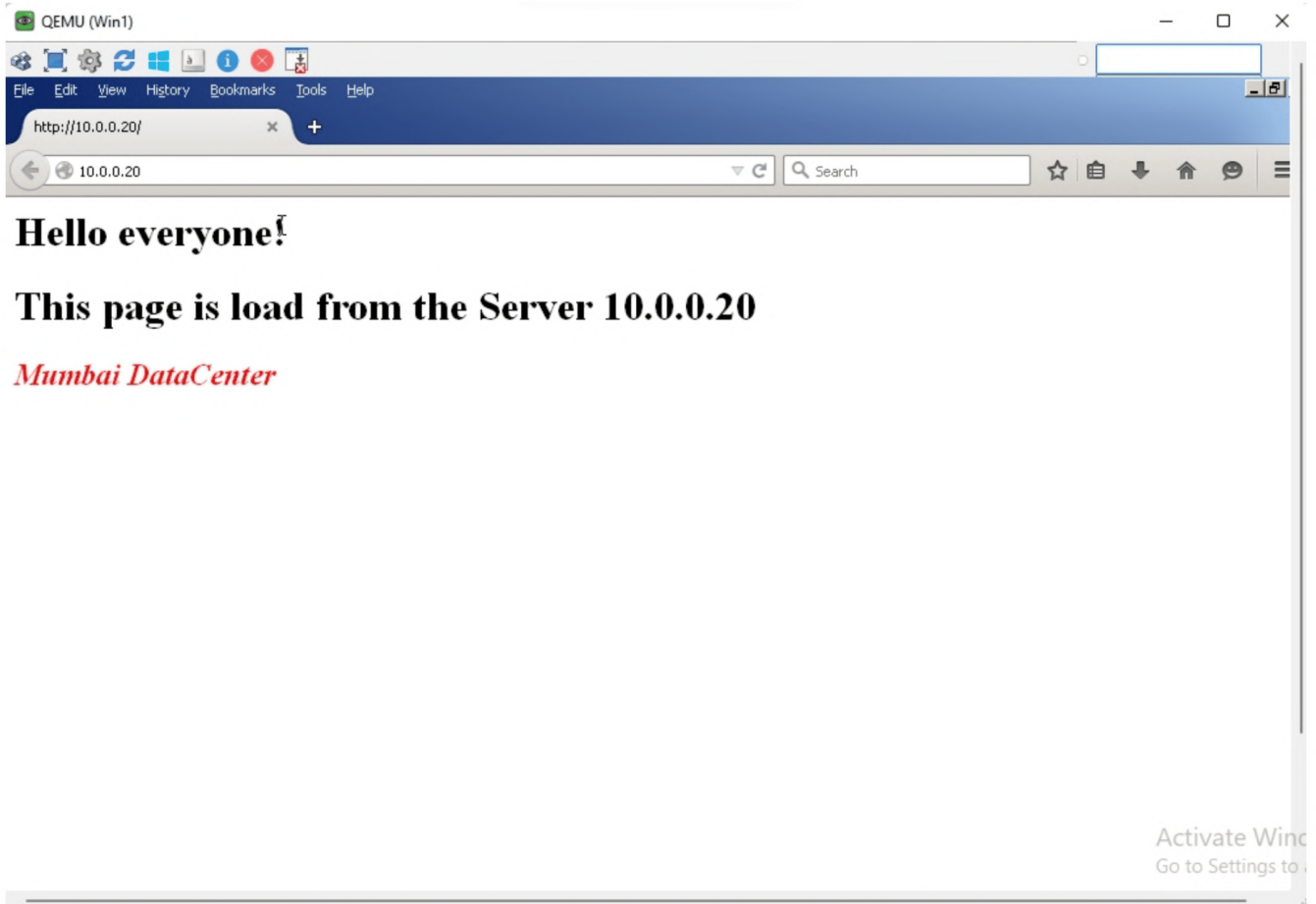


Fig. 23 - Browser test confirming Mumbai-DC web server via IP address.

24. DNS Record for Mumbai-DC

In DNS Manager on the Global-DNS server, a new Host (A) record 'www' is created under the 'mumbaidc.com' forward lookup zone, mapping www.mumbaidc.com to IP address 10.0.0.20.

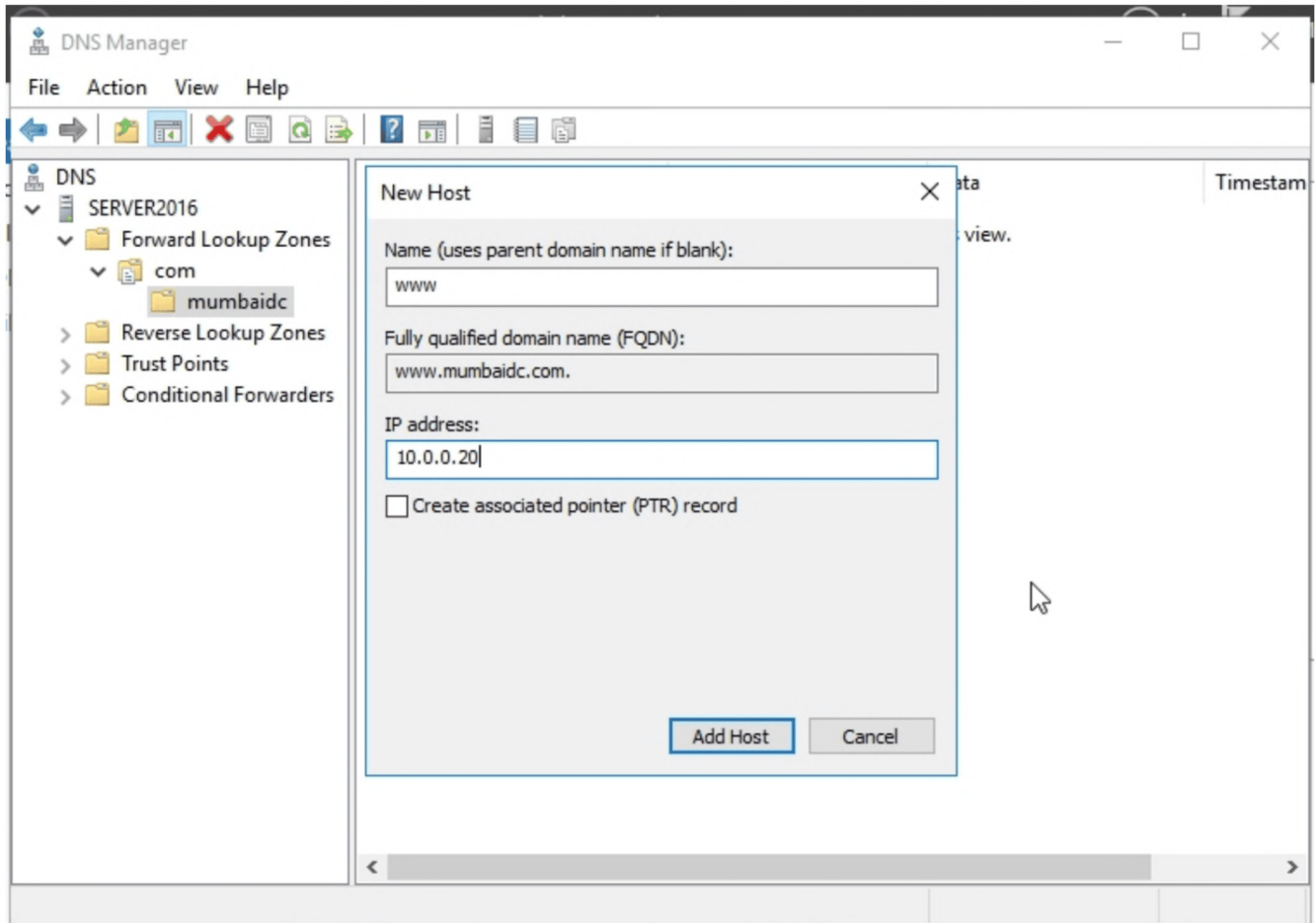


Fig. 24 - Creating the www.mumbaidc.com A record in DNS Manager.

25. DNS Record for Chennai-DC

Similarly, a Host (A) record 'www' is created under the 'chennaiidc.tn' zone, mapping www.chennaiidc.tn to the Chennai-DC server's IP address, 200.10.156.30.

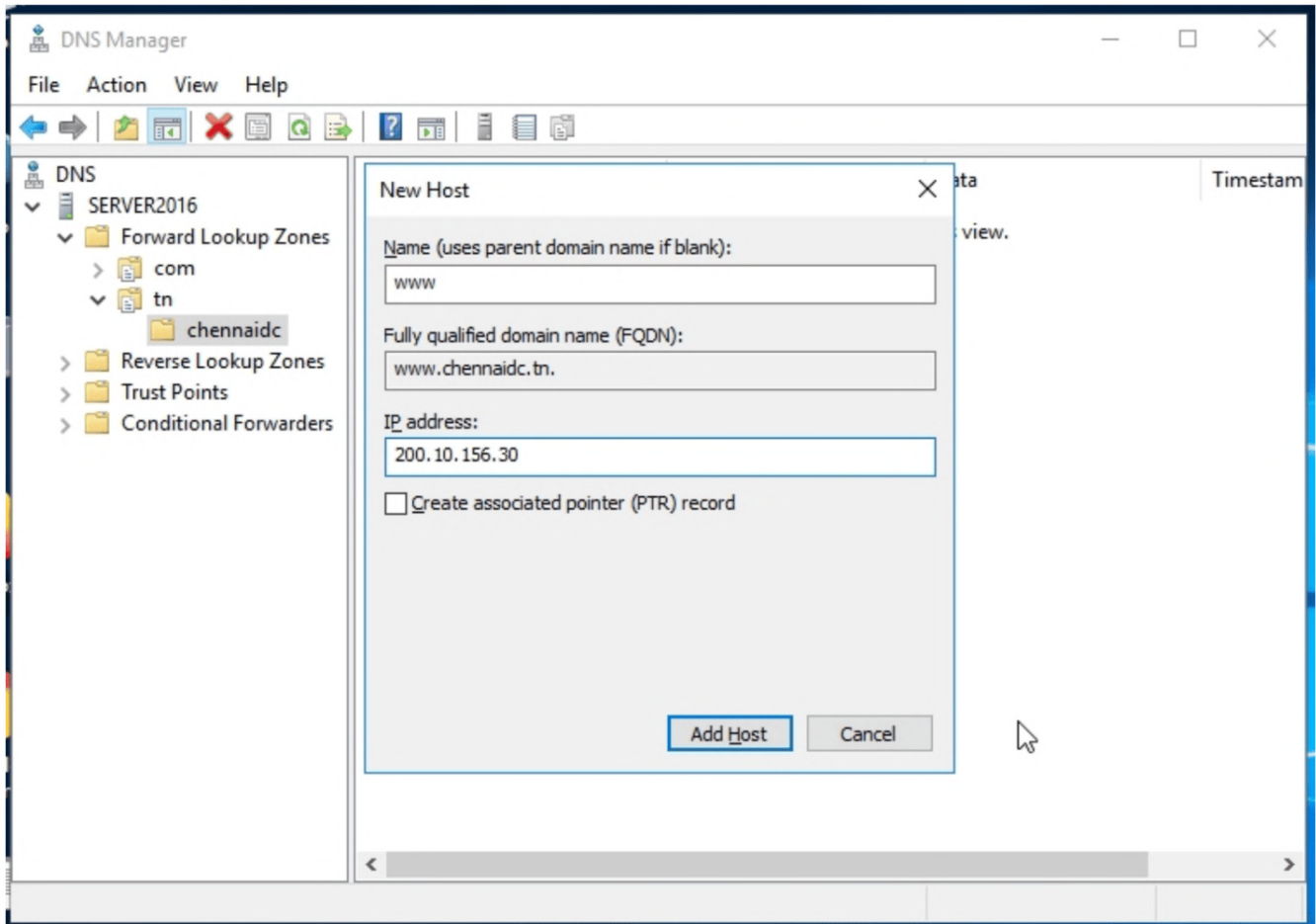


Fig. 25 - Creating the www.chennaic.tn A record in DNS Manager.

26. DNS Record for Pune-DC

A Host (A) record 'www' is created under the 'punedc.in' zone, mapping www.punedc.in to the Pune-DC IIS server's IP address, 182.46.0.10.

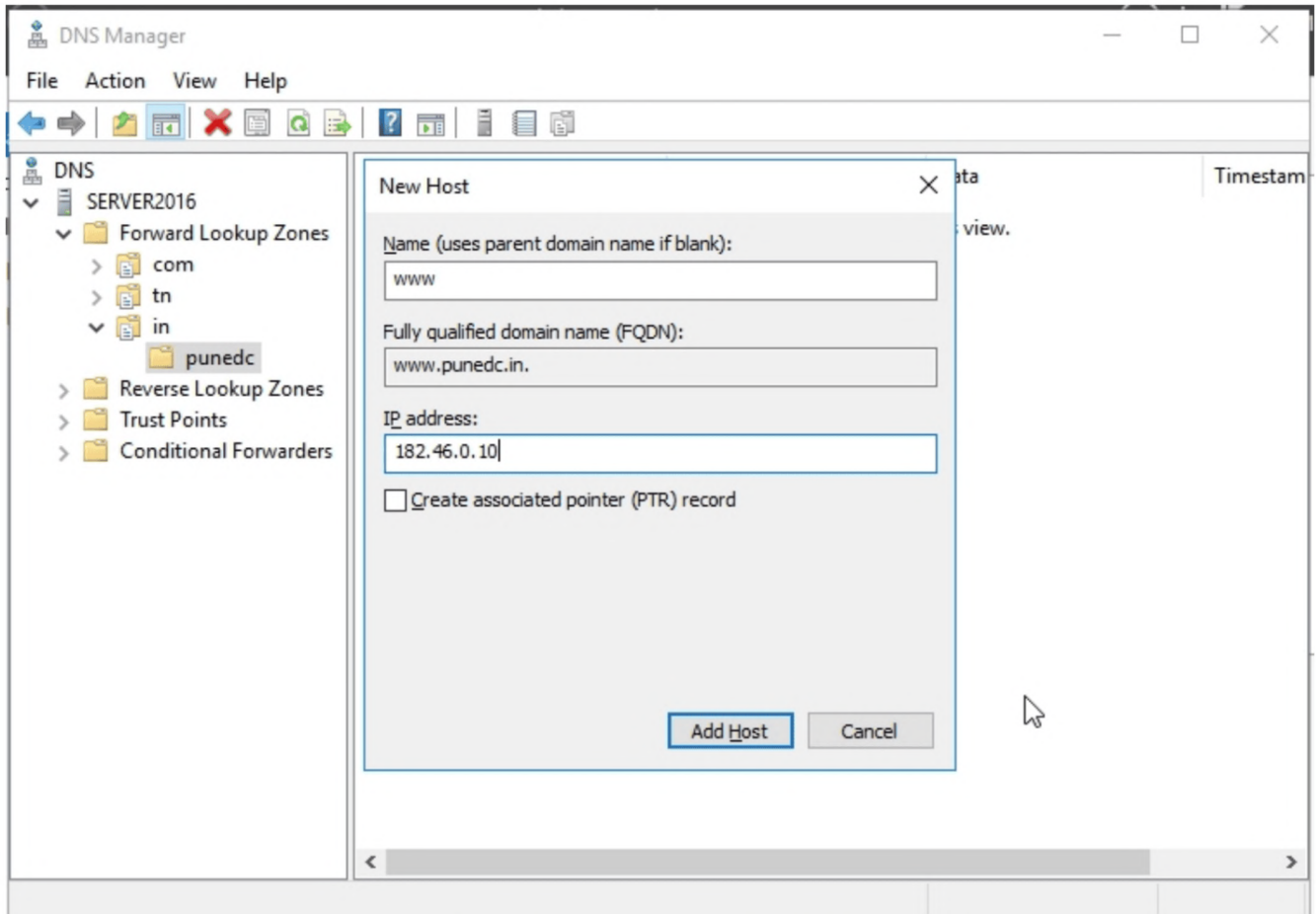


Fig. 26 - Creating the www.punedc.in A record in DNS Manager.

27. DNS Resolution Test - Mumbai-DC

From the branch client, browsing to <http://www.mumbaidc.com> successfully resolves via DNS and loads the Mumbai-DC page, proving the DNS record and name resolution are working correctly.

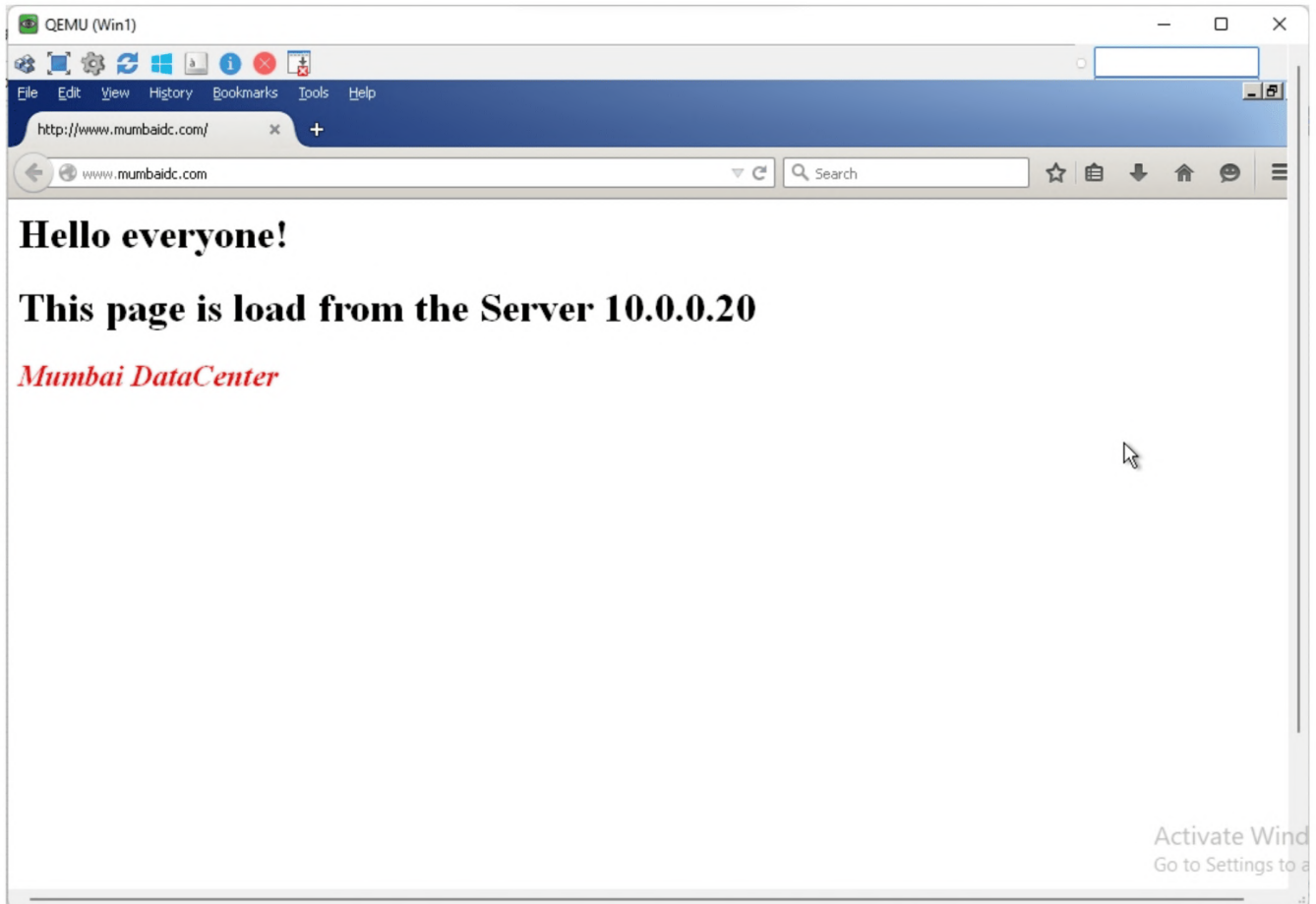


Fig. 27 - Successful DNS-based access to www.mumbaidc.com.

28. DNS Resolution Test - Chennai-DC

Browsing to <http://www.chennaiidc.tn> resolves via DNS and loads the Chennai-DC page, confirming the Chennai DNS record is functioning correctly.

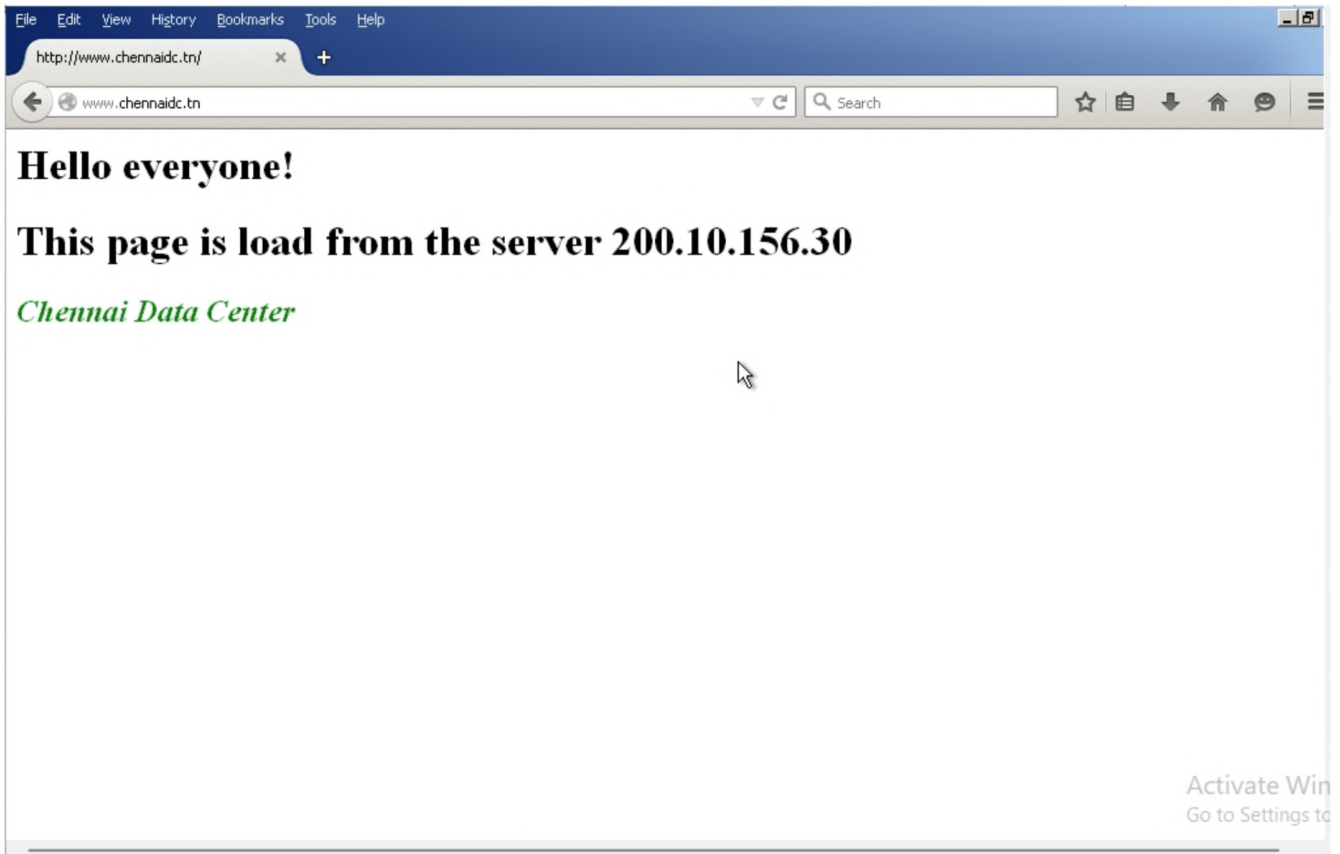


Fig. 28 - Successful DNS-based access to www.chennaiidc.tn.

29. DNS Resolution Test - Pune-DC

Browsing to <http://www.punedc.in> resolves via DNS and loads the default IIS welcome page, confirming the Pune-DC DNS record works and completing the end-to-end DNS verification for all three data centers.

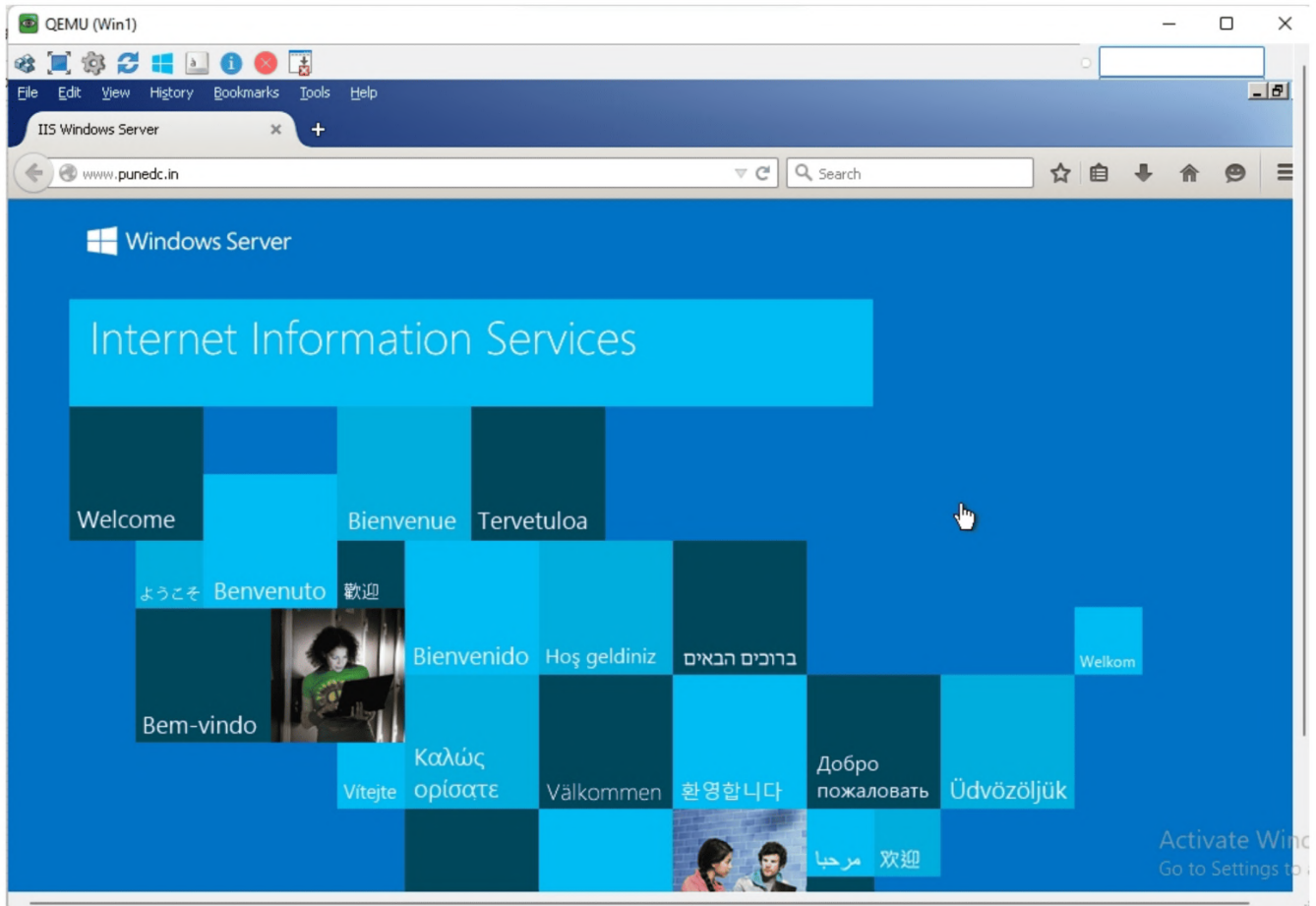


Fig. 29 - Successful DNS-based access to www.punedc.in.